



Ap/Kp and F10.7 Forecasting

**45-Day Product User
Engagement
January 21, 2026**

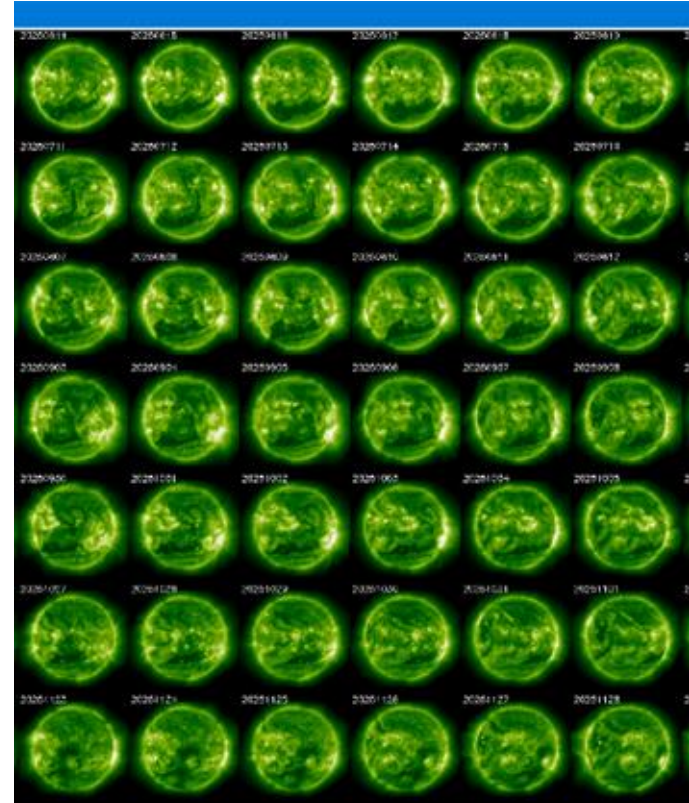
Overview

- Factors Affecting Ap/Kp
 - Coronal Hole High Speed Streams (CH HSSs)
 - Co-rotating Interaction Regions (CIRs)
 - Coronal Mass Ejections (CMEs)
- Kp Network - Kennziffer meaning “characteristic digit”
- Forecasting Kp/Ap - Ap is the planetary amplitude
- F10.7 Data Ingest - Indicator of solar activity measured at 2800 MHz
- F10.7 Forecasting
- Forecasting Challenges

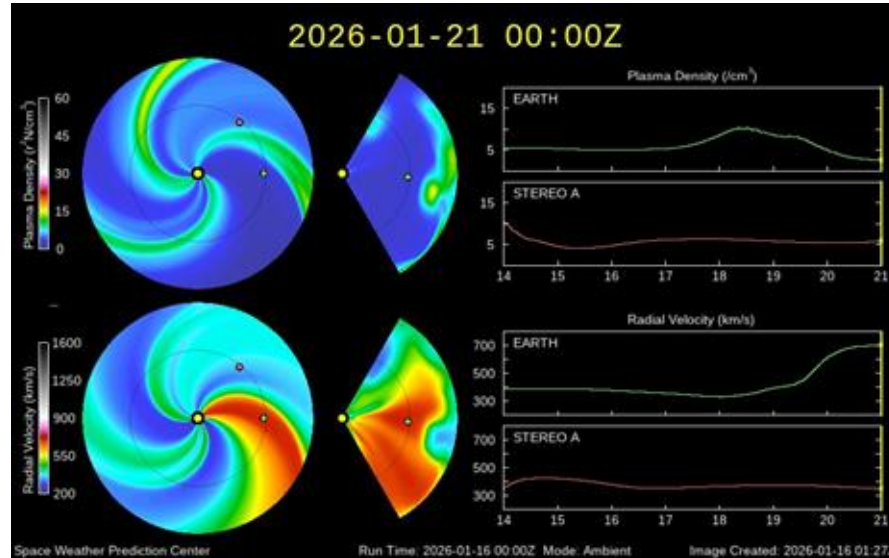


Factors Affecting Ap/Kp (CH HSS)

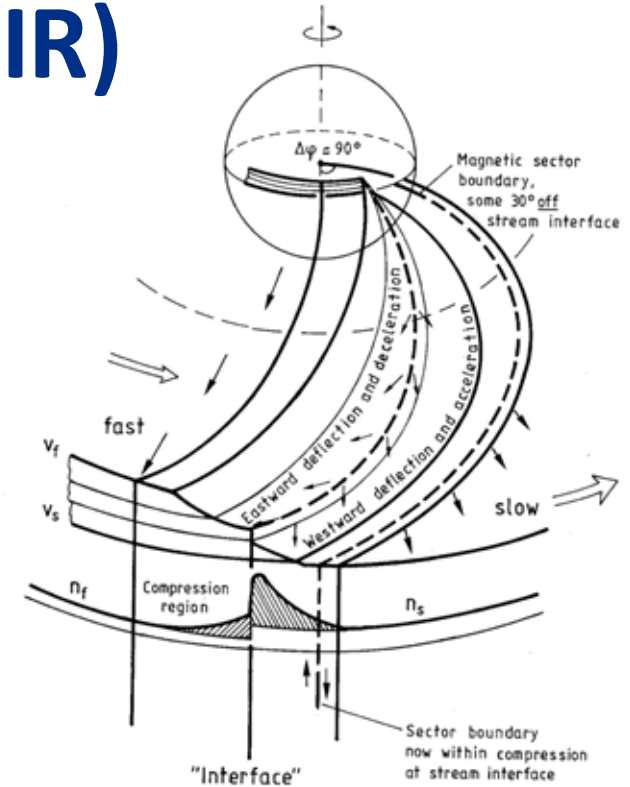
- Coronal Holes correspond to areas of “open” magnetic fields on the Sun. They are lower in density
- The solar wind emanating from these coronal holes is high in velocity (~500+ km/s).
- Generally smaller, closer to the equator in solar maximum.
- Seasonal differences in Ap/Kp values.
- Can be forecasted through observations, modelling, and recurrence.



Factors Affecting Ap/Kp (CIR)



- Co-rotating Interaction Regions are areas of compressed plasma between slow moving solar wind and high speed streams.
- Like CH HSS these can be recurrent events.
- Kp values usually in the 4 to 6 range with some reaching as high as 7.
- Ap is issued on a 24 hr time frame. Splitting days can change values.



Schwenn, R., 1990, "Large-Scale Structure of the Interplanetary Medium", in Physics of the Inner Heliosphere, Vol. I: Large-Scale Phenomena, (Eds.) Schwenn, R., Marsch, E., vol. 20 of Physics and Chemistry in Space, pp. 99–181, Springer, Berlin; New York.

Factors Affecting Ap/Kp (CMEs)

- Coronal Mass Ejections (CMEs) are a large outflow of plasma from the Sun and are often associated with erupting prominences, disappearing solar filaments structures, and/or flares.
- Can contain billions of tons of matter accelerated to several million miles per hour.
- Modelling challenges include multiple CME interactions, limited viewpoints, and unknown magnetic field characteristics until it reaches spacecraft at L1.
- Modelling can help with timing and approximate speed at Earth.
- Right is model of Veterans Day Storms 11-13 Nov. 2025.

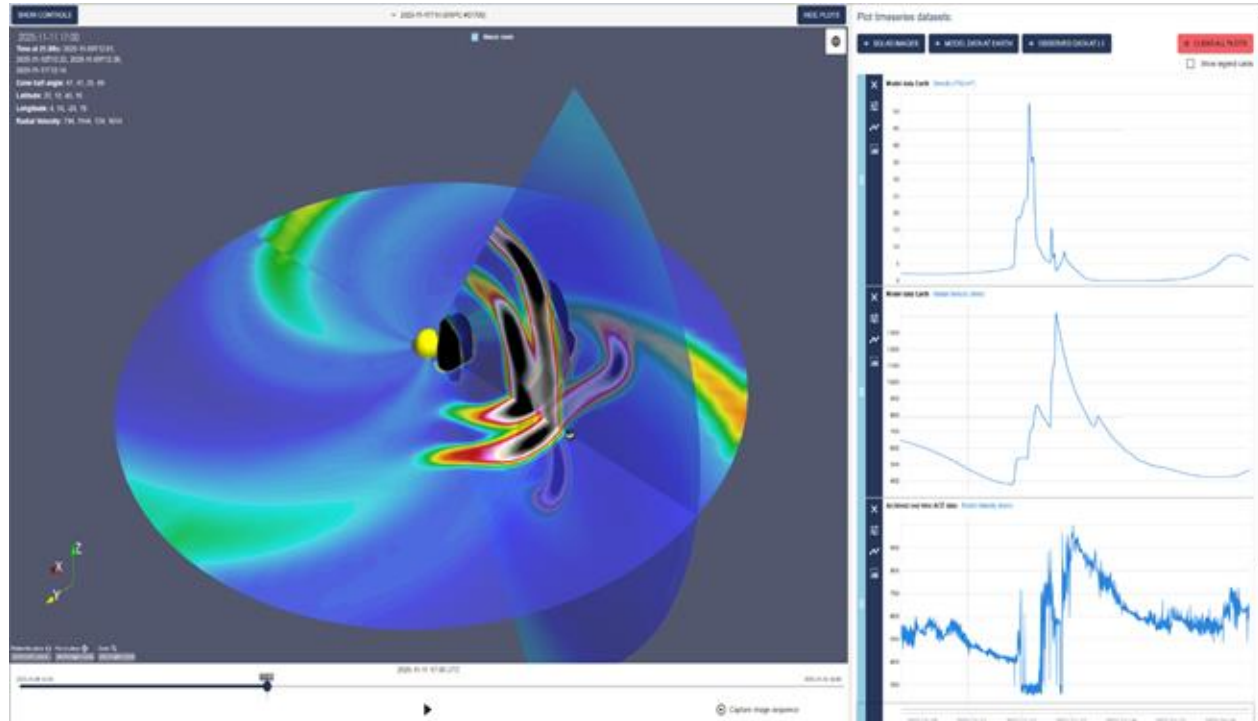
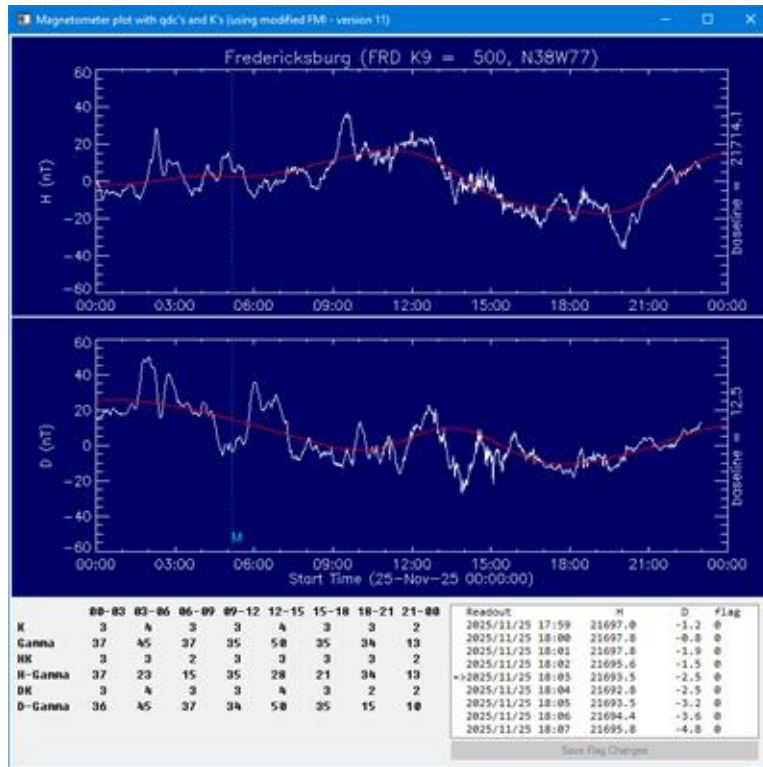
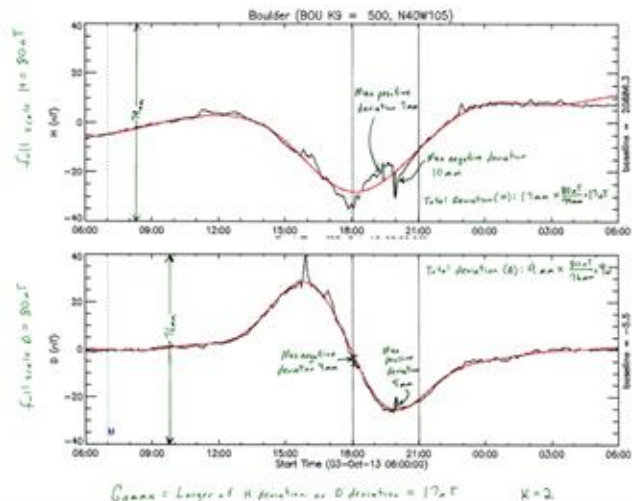


Image Courtesy University of Colorado Laboratory for Atmospheric and Space Physics at <https://swx-trec.com/h3lioviz/>



Kp Network

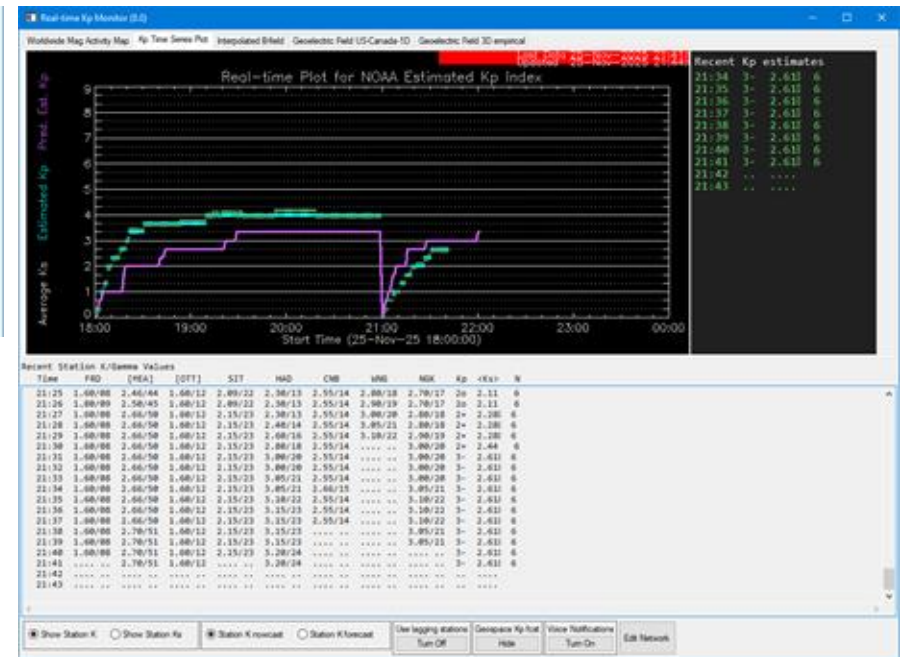
- Pre-2012: Boulder Magnetometer used for Alerts.
- 2012+: SWPCs estimated Kp network uses 8 magnetometer stations across Europe, North America, and Australia.
- Deviations from an estimated quiet day curve is measured in nano-Teslas (nT) and a station K value is computed based on its latitude.



Station K and nT deviation equivalent at Boulder, Co.

K	nT	a-index
0	0-5	0
1	5-10	0
3	10-20	7
2	20-40	15
3	40-70	15
4	70-120	15
27	120-200	15
5	200-330	15
48		15
6		15
80		15
7		15
140		15

Kp Network



- The stations are averaged together to determine the Ks value every minute.
- Kp values are set in 3 hr increments beginning at 0000z.
- Ap index is a 24 hour index derived from the K indices.
- Each 3 hourly K value is converted back into a linear scale (the a-index). The daily Ap value is an average of all the 3 hourly a-indices.



Forecasting Kp/Ap

- Climatological forecast: is the average conditions over time (benchmark).
- Persistence forecast: tomorrow's condition will mimic today's.
- Recurrence forecast: max Kp level observed 27 days ago (Approx. 1 full Solar rotation).
- **SWPC forecasts** three days of Kp values and seven days of Ap values. A 3-day forecast goes out twice a day at 0030z and 1230z.
- **Longer predictions** are included in the 27-day Outlook product that gets issued once a week on Sunday night (MST).

7 Day Geomagnetic Forecast 25-Nov-2025 21:53:48 (Forecaster: forecaster)

Climatology						
Q	U	A	nS	MS	SE	<A>
64	20	9	4	2	1	13

Persistence						
Q	U	A	nS	MS	SE	<A>
0	0	100	0	0	0	27

Recurrence Forecast							
Date	Q	U	A	nS	MS	SE	<A>
26	0	0	0	100	0	0	37
27	0	0	0	100	0	0	24
28	0	100	0	0	0	0	13
29	0	0	100	0	0	0	18
30	0	0	100	0	0	0	20
1	0	100	0	0	0	0	10
2	0	0	0	0	100	0	27

Physical Forecast							
Date	Q	U	A	nS	MS	SE	<A>
26	100	0	0	0	0	0	5
27	100	0	0	0	0	0	5
28	100	0	0	0	0	0	5
29	100	0	0	0	0	0	5
30	100	0	0	0	0	0	5
1	100	0	0	0	0	0	5
2	100	0	0	0	0	0	5

Guidance Forecast							
Date	Q	U	A	nS	MS	SE	<A>
26	77	5	2	14	0	0	12
27	77	5	2	14	0	0	10
28	77	19	2	1	0	0	9
29	77	5	16	1	0	0	9
30	77	5	16	1	0	0	9
1	77	19	2	1	0	0	8
2	77	5	2	1	14	0	10

☐ Show Three Hour Probabilities
☐ Show Three Hour Kforecast

Weights						
Phy	Rec	Per	Cli			
9	2	0	4			
9	2	0	4			
9	2	0	4			
9	2	0	4			
9	2	0	4			
9	2	0	4			

(Enter values from 0 to 9)

Forecaster Forecast							
Date	Q	U	A	nS	MS	SE	<A>
26	0	10	20	40	25	5	36
27	0	9	35	40	15	1	30
28	19	35	20	20	5	1	25
29	19	35	20	20	5	1	12
30	19	35	20	20	5	1	8
1	53	25	15	5	1	1	5
2	53	25	15	5	1	1	5

3-Day Kp-index Forecast									
Date	03	06	09	12	15	18	21	24	<A>
26	5-	4+	4+	5-	5-	5-	4+	4+	36
27	5-	4+	5-	40	4-	4-	40	4+	30
28	40	40	4-	4-	4-	40	40	4-	25

☒ Kp4(Kp) ☐ Ap4(Kp)

Summary

01 Geomagnetic Activity Probabilities

26 Nov-28 Nov

a. Planetary

Active: 20/35/20

Minor Storm: 40/40/20

Moderate Storm: 25/15/05

Strong-Extreme Storm: 05/01/01

F10.7 Data Ingest

- Official F10.7cm (2800 MHz) solar flux unit (sfu) values measured at 1800z, 2000z, and 2200z.
- Radio flux values are received daily from Canada's Dominion Radio Astrophysical Observatory in Penticton, Canada.
- Other radio frequencies are monitored and received daily from US Air Force's Radio Solar Telescope Network (RSTN) partners.
- Corrections to F10.7 cm number can be estimated from RSTN Number.

Solar Indices Monitor

2323 UTC 25 Nov 25

Magnetic Indices

24 Nov 25

25 Nov 25

	03	06	09	12	15	18	21	24	A	03	06	09	12	15	18	21	24	Running A
Fredericksburg	3	2	2	3	4	3	3	3	15	3	3	4	3	4	3	3	*	18
Boulder	3	2	3	3	5	3	3	3	18	4	4	4	5	5	4	3	*	29
College	1	3	3	5	6	6	6	4	44	4	4	6	7	6	6	4	*	61
Planetary - NOAA	3+	3o	3+	4+	4+	4o	4+	3+	22	5-	4+	5o	4o	4+	4+	4o	*	32
Planetary - AF	3	3	3	4	4	4	4	3	22	5	4	5	4	4	4	4	*	32

Radio Flux Values ($10^{-23} \text{Wm}^{-2} \text{Hz}^{-1}$)

25 Nov 25

Station / CAMP	245	410	610	1415	2695	4995	8800	15400
Leameth / 8323 UTC	35	37	55	*	*	*	*	*
San Vito / 1844 UTC	13	28	49	86	107	130	278	573
Sag Har / 1721 UTC	18	31	39	70	114	111	280	518
Pahoa / 2208 UTC	21	39	62	84	107	162	258	521
Penticton 10.7 cm	Morning	115	Noon	116	Afternoon	117		

Daily Parameters

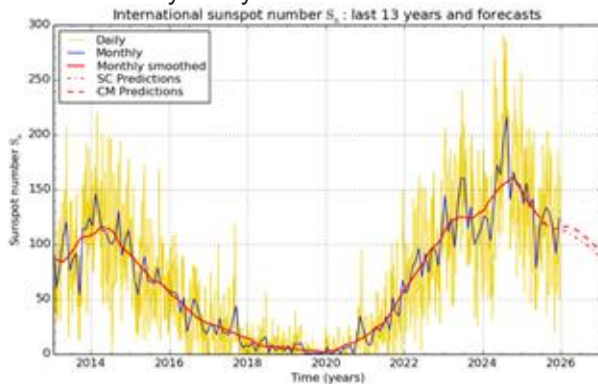
24 Nov 25

Event totals	C: 5	M: 0	X: 0	Type II	0	Noon 10cm flux	116	GOES 18 X-ray bkgd	B3.9
Sunspot number	99	Area	230	New Groups	1	GOES 18 electron fluence 2 MeV	1.30e+07		
GOES 18 proton fluence 1 MeV	5.75e+06	10 MeV	1.49e+04						

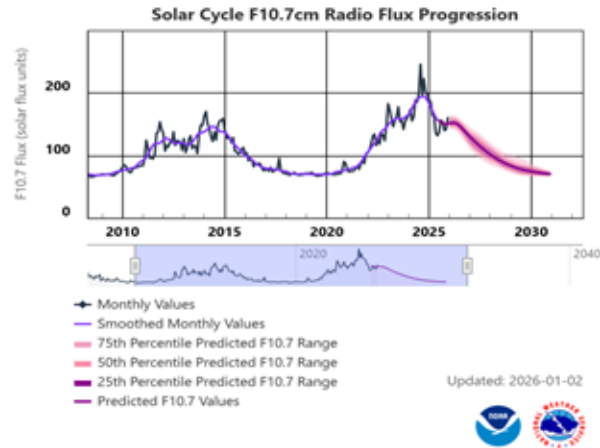
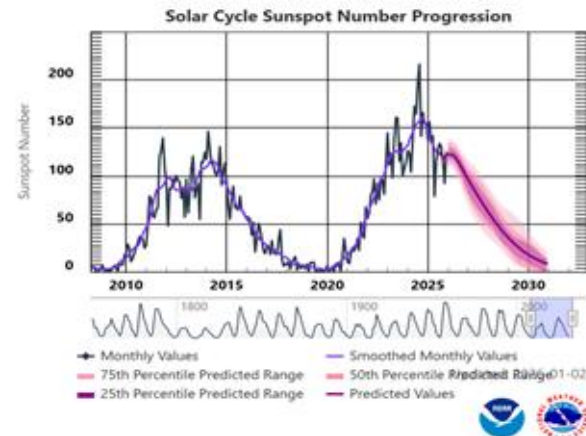


F10.7 Forecasting

- There is a relationship between F10.7 flux and Sunspot numbers.
- Prediction for a solar cycle is conducted by a panel from NOAA, NASA, and International Space Environment Services (ISES) representatives.
- SWPC plots on the right are showing smoothed monthly values and predicted values.
- Sources:
 - SWPC Daily Sunspot Numbers
 - Wolf Number (relative sunspot number) ($R = k(10g + s)$)
 - k = factor that varies with observers (usually 1)
 - g = Number of sunspot groups
 - s = Number of individual spots
 - Solar Influences Data Center (SIDC) Brussels
 - Penticton, B.C., Canada
- Bottom plot is from SIDC's Sunspot Index and Long-term Solar Observations (SILSO).
- Daily Sunspot Numbers can vary wildly.



SILSO graphics (<http://sidc.be/silso>) Royal Observatory of Belgium 2026 January 1

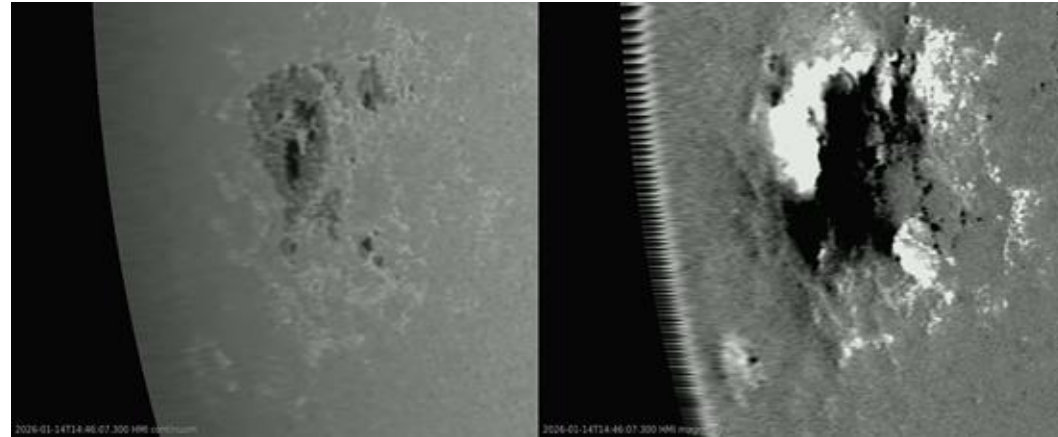


Updated: 2026-01-02

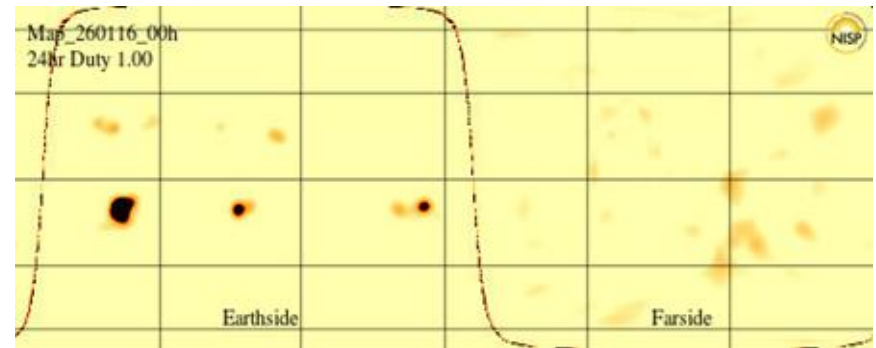


F10.7 Forecasting

- F10.7 forecasting is closely related with sunspot regions currently on the Earth-facing solar surface.
- Sunspots occur at areas where magnetic field lines emerge from the interior of the Sun.
- Forecasting F10.7 values involve looking at:
 - Evolution of Sunspot groups
 - Rotation on/off the Sun of these groups.
 - Potential return of larger groups.
- NSO/GONG Helioseismology can provide an idea if there is the development of a larger active region.



Video is a 2-day movie loop of Regions 4341, 4345, and 4346 showing an example of Sunspot evolution (Courtesy NASA/ESA Jhelioviewer). Below is a recent far-sided image using helioseismology (Courtesy NSO/GONG).



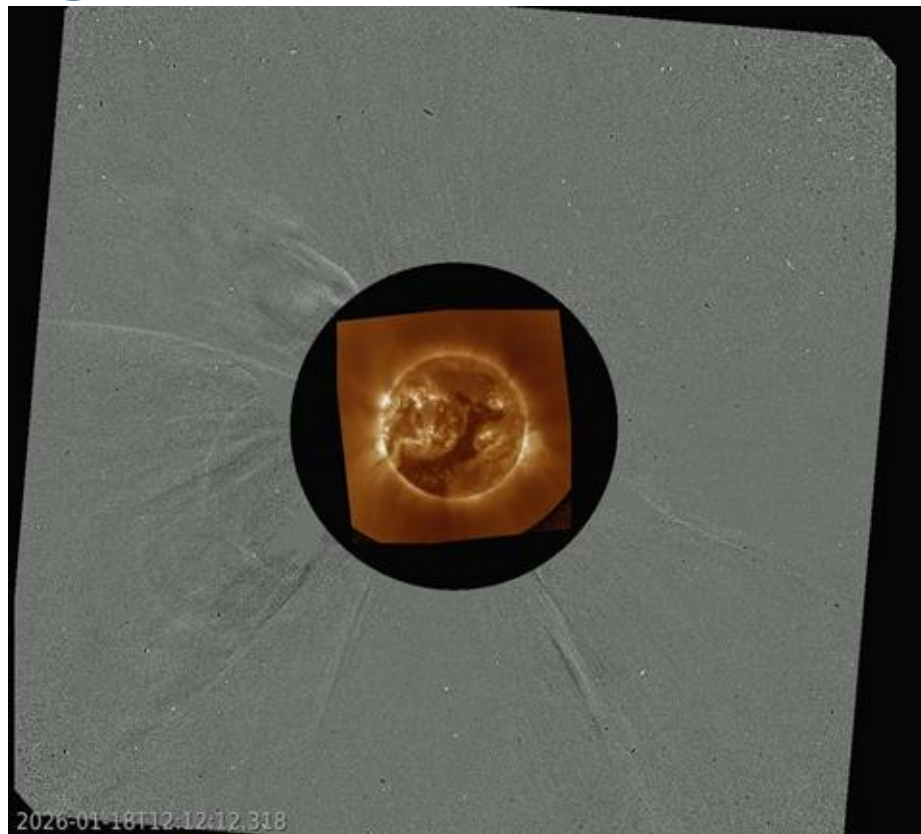
Forecasting Challenges

- Data degradation due to space weather effects.
- Data quality.
- Sun-Earth line only provides one frame of reference. L5 observations needed.
- Uncertainty in timing of events.
- Multiple CME/CH HSS interactions.
- CME magnetic field unknown until it reaches L1.
- 24 Hr set windows for daily Ap values.
- Sunspot variability.

Video is of the recent X1.9 flare at 18:09 UTC on 18 Jan. from Region 4341.

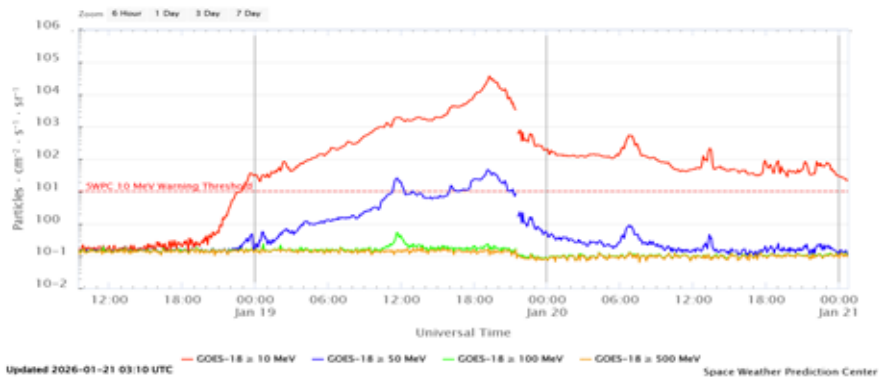
A proton event reached the S1 (Minor) levels at 22:55 UTC on the 18th and reached a max of 37,000 pfu (S4 - Severe) at 19:15 on 19 Jan.

Higher level proton enhancements can cause problems for optics and measurements.

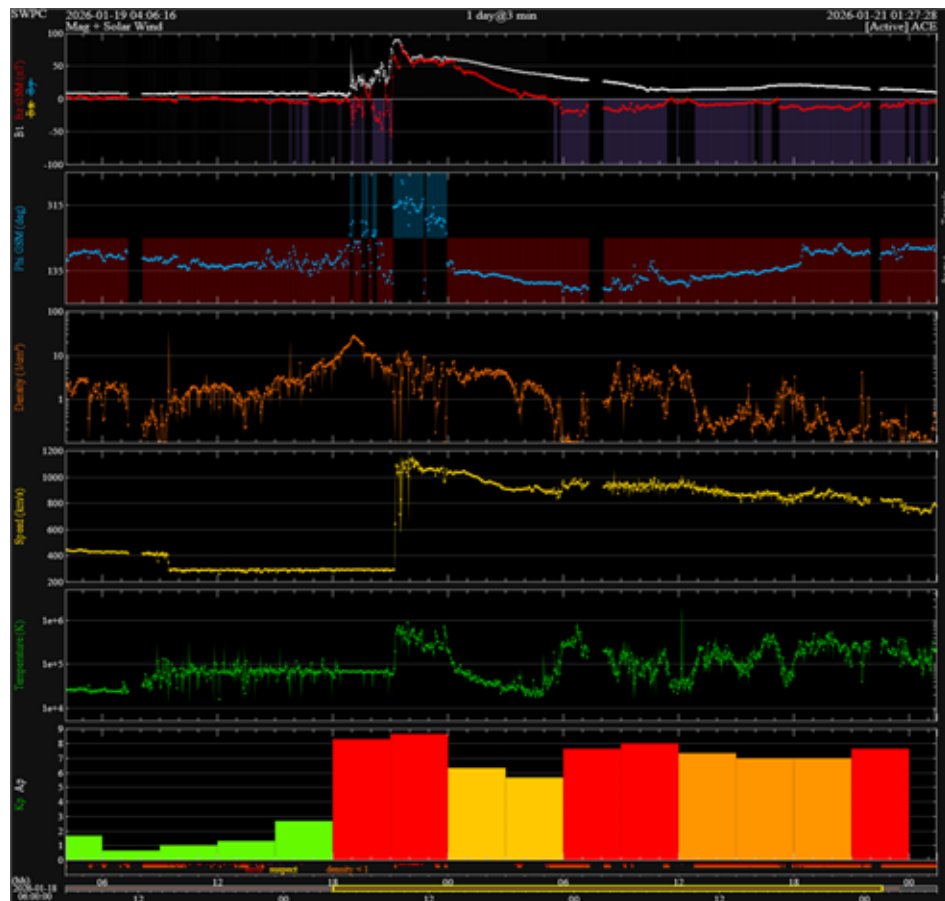


Challenges

GOES Proton Flux (5-minute data)



- Data dropouts like in solar wind data on the right are caused by higher energy proton enhancements.
- SWFO-L1 initial measurements seem more robust.



Ap/Kp and F10.7 Forecasting

Any questions?

